

ELFS : Compression Focused File System

ELFS, also pronounced as elves, is a modern compression based real time r/w file system.

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Abstract - Elfs is a command-line middleware file system designed for efficient compression. It builds upon standard file systems like ext3, ext4, NTFS, and Fat32 but emphasizes a compression-centric approach. Inspired by SquashFS and Dwarfs, Elfs is tailored for applications requiring optimized storage solutions. The system's goal is to develop a read/write file system that functions seamlessly on a compressed drive. Acting as an intermediary, the Elfs-daemon ensures smooth access without user intervention. Unlike conventional file systems, Elfs integrates compression directly into its structure, maximizing storage efficiency while maintaining performance. With its advanced compression techniques, Elfs provides an ideal solution for applications needing effective data management. Its potential use cases include software distribution, cloud storage, and embedded systems, where space conservation and seamless access are crucial.

In the age of AI, a massive amount of data is being used and generated daily. There is a growing need for solutions where compressed data can be used just like normal data, without incurring the cost of decompression upfront. ELFS solves this problem by creating a system where data is transparently managed between compressed and uncompressed states, ensuring users can access compressed files seamlessly without performance degradation

Keywords—Elfs, Compression, Zstd, File System.

I. INTRODUCTION

With the rapid increase in data generation and storage requirements, the demand for more efficient storage solutions has become a critical challenge. Traditional file systems, though reliable, often struggle to provide effective data compression while maintaining fast access speeds. **Elfs** aims to solve this problem by integrating seamless real-time compression and decompression capabilities. The **Elfs-daemon** runs in the background, handling file compression

efficiently while ensuring uninterrupted data access.

Several references highlight key aspects related to file system performance and compression techniques. For instance, reference [4] examines the difficulties of achieving real-time compression while balancing efficiency and low-latency performance. Reference [6] explores advanced compression methods to enhance data storage and throughput. A comparative analysis of modern file systems such as ext4, Btrfs, and ZFS is provided in [7], while [8] discusses strategies to ensure predictable performance in real-time file systems through scheduling and consistency mechanisms. Modern compression techniques for optimizing storage efficiency and data transfer speeds are reviewed in [9]. Performance assessments of contemporary file systems, focusing on speed, reliability, and scalability across different conditions, are covered in [10]. Additionally, strategies for achieving consistent real-time performance, particularly in latency-sensitive environments, are detailed in [11]. The development of specialized file systems designed to improve compression efficiency is highlighted in [26], which introduces Lecramfs, a system optimized for flash-based storage in portable devices. Enhancements to mobile device responsiveness through quasi-asynchronous I/O operations are discussed in [27], improving real-time application performance. For embedded systems, reference [30] presents an Execute-In-Place (XIP) architecture that minimizes memory usage and accelerates boot times, leading to better system performance in constrained environments. Reference [31] proposes a unified approach combining non-volatile memory (NVM) and NAND flash, improving smartphone storage in terms of speed, reliability, and cost-effectiveness. Furthermore, reference [32] delves into the design and efficiency of distributed file systems, emphasizing scalability and data management across multiple locations. A foundational principle in modern storage systems, the RAID concept, is introduced in [33], demonstrating how multiple inexpensive disks can be used to enhance data redundancy, speed, and reliability. Meanwhile,

Lecramfs is further examined in [42] for its role in optimizing compression for flash-based storage in portable devices, addressing mobile system constraints. Efficient data retrieval and indexing techniques are explored in reference [45], which reviews D. Comer’s 1979 paper on B-trees. This study discusses the significance of B-tree structures in handling large datasets, particularly in databases and file systems, and highlights variations like B+ trees, commonly used in disk-based storage systems.

Btrfs [1], a modern file system with flexible features, supports compression but falls short in terms of real-time decompression efficiency for high-performance applications. SquashFS and Dwarfs [2][3] are designed for read-only compressed file systems but lack support for real-time decompression or live system mounting. Additionally, reference [3] analyzes the application of various compression algorithms in file systems, outlining the trade-offs between compression ratio and system performance. Finally, Anderson [4] investigates the complexities of real-time file system compression, emphasizing the challenge of maintaining both speed and efficiency.

II. LITERATURE SURVEY

The concept of compression-based file systems has been explored in numerous studies, with several solutions focusing on improving compression ratios without sacrificing performance. A review of reference papers reveals that:

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This literature survey highlights several ongoing challenges in file system design and performance. Feature-rich systems like Btrfs add complexity, which can make them harder to maintain and sometimes slower. Compression helps save space but often increases CPU use and delays, especially in mobile and real-time systems. Real-time file handling is difficult due to the need for quick response and low latency, which current designs

struggle to provide. In mobile and embedded environments, performance is limited by inefficient operations and slow flash storage.

Gaming and media applications need fast file access, but many file systems aren’t built for that level of speed. Network-based systems like NFS and CIFS also face issues with performance and scalability. Even though distributed file systems improve data access and reliability, they still have trouble managing very large setups. Emerging technologies like qNVRAM and Execute-In-Place offer improvements, but bringing them into current systems isn’t easy. Overall, finding the right balance between speed, storage space, and reliability continues to be a major challenge.

PROBLEM STATEMENT

To overcome these problems, several solutions can be applied. Hybrid file systems can adapt to different workloads for better efficiency. Optimized compression algorithms can balance speed and storage capacity. Hardware acceleration can reduce delays in real-time compression. Standardized benchmarking tests can make performance comparisons easier. Smarter caching and scheduling strategies can enhance both speed and reliability. More accurate profiling tools can improve storage behavior analysis. Advanced error correction techniques can strengthen RAID redundancy. Optimizing distributed algorithms can help shared-disk systems scale more effectively. Automated tuning can adjust file systems based on workload demands. Finally, reducing unnecessary journaling can improve performance in systems that track file changes.

III. METHODOLOGY

The methodology behind Elfs: Efficient Lossless File System is straightforward yet powerful, focusing on seamless integration between *compressed* and *uncompressed* data. The key concept involves two types of storage blocks:

1. Cold Storage Blob : A Compressed Storage Block
2. Hot Storage Blob : An Uncompressed or active Storage Block.

The Elfs-daemon plays a central role in this system. It operates as a real-time daemon that works in tandem with the Linux kernel. When a user accesses a file on a compressed partition, Elfs-daemon automatically decompresses the file in the background. The decompression process occurs dynamically, ensuring that users can interact with the file system as if the data were uncompressed, without needing to manually manage compression and decompression operations. This background operation consumes *minimal* CPU and RAM resources, ensuring high performance and low overhead.

The implementation of Elfs: Efficient Lossless File System is built around three main applications, all written in Rust. Rust's performance, memory safety, and concurrency features make it an ideal choice for building a high-performance file system that interacts seamlessly with cloud and distributed storage systems.

The architecture of Elfs is designed to be simple yet effective:

- a) elfs (Command Line Tool): A utility that can be used to create a file system on a compressed drive.
- b) elfs-cat: A utility designed to allow users to read and print specific files stored in the compressed (cold) storage. This demonstrates how a file can be accessed from the compressed partition and decompressed as needed.
- c) elfs-daemon: The core real-time daemon responsible for decompressing files as they are accessed. It translates requests for compressed storage into readable, uncompressed data on the fly.

This architecture allows Elfs to seamlessly integrate into existing workflows while providing the advantages of compression without sacrificing user experience or system performance.

The three core applications in the Elfs ecosystem are:

- a) elfs: This command-line tool is used to create a file system on a compressed drive using the ``mkfs`` command.
- b) elfs-cat: A utility designed to allow users to read and print specific files stored in the compressed (cold) storage. This demonstrates how a file can be accessed from the compressed partition and decompressed as needed.
- c) elfs-daemon: The key component that operates in real-time, continuously monitoring file accesses. It decompresses the files in the background as they are accessed, ensuring a seamless user experience.

The system architecture is focused on integrating with cloud and distributed storage systems. When data is written, it is initially stored as *cache blobs* (uncompressed data). As the system detects that these blobs are no longer being accessed (i.e., "cold"), it compresses them into *cold storage blobs* and replaces the cache blobs. This dynamic management ensures that frequently accessed data remains readily available in its uncompressed form, while less accessed data is efficiently compressed to save storage space.

By using Rust and focusing on distributed and cloud storage, Elfs ensures that it can scale efficiently for a variety of storage environments, providing high compression without compromising performance.

The core compression algorithm used in elfs is Zstandard (zstd), developed by engineers at Facebook. Zstd is known for its high compression ratios and extremely fast compression and decompression speeds, making it an ideal choice for a file system that requires real-time, efficient data handling. Access speed is a critical factor in file systems, and Zstd strikes an excellent balance between speed and compression, ensuring that data can be accessed quickly while being stored efficiently.

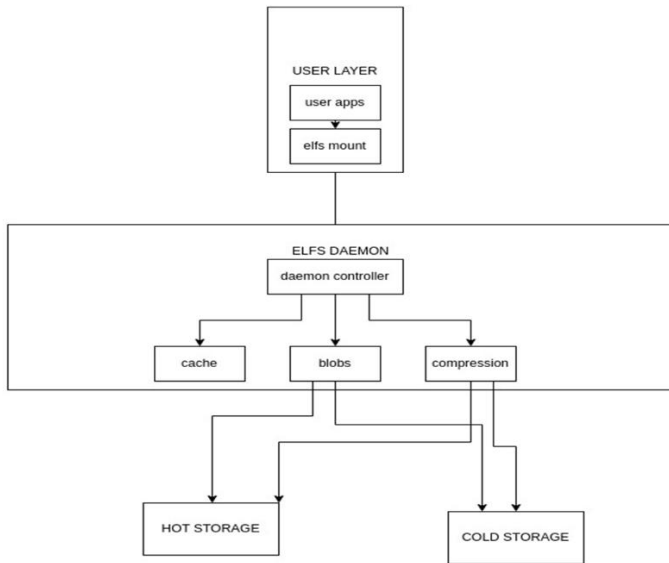


Fig:3.1- Architecture of ELFS

- Compression Performance

We conducted several tests to evaluate the compression performance of zstd in comparison to other algorithms like xz, gzip, and zip Below are the results for compressing a large container file (*rust_ros2_container.tar*).

- require frequent access.
- XZ, which took the longest time 25 minutes 39 seconds to compress, also resulted in a reasonably small file size 751.19 MB, but its slower performance makes it less suitable for a real-time system like Elfs.
 - Gzip and Zip both performed well in terms of speed, taking around 2-3 minutes for compression, but their compression ratios are inferior to that of Zstd, resulting in larger file sizes (1.28 GB).

Here is a graph that demonstrates the relationship between compression ratio and decompression speed. As the compression ratio increases, the decompression speed tends to decrease, which is common in many lossless file systems. This trade-off is crucial when designing efficient file systems, as higher compression ratios can reduce storage space but may slow down data retrieval.

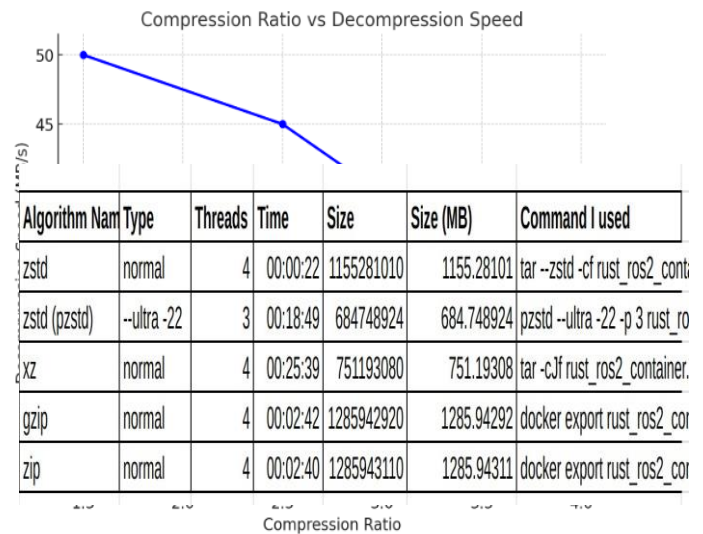


Fig:4.1 – Graph showing Compression ratio vs Decompression speed

IV. RESULTS

- zstd(normal) achieved a compression speed of 22 seconds and resulted in a file size of 1.155 GB. This compression speed and efficiency make it well-suited for use in Elfs, where real-time decompression is a priority.
- Zstd (pzstd), using the `--ultra -22` compression option, took 18 minutes 49 seconds but achieved a smaller file size (**684.75 MB**). While the compression time is significantly longer, the smaller size could be beneficial for cold storage blobs that do not

V. CONCLUSION

Zstd Emerges as the most optimal choice for Elfs due to its combination of speed and good compression. It provides fast compression and decompression while ensuring that the file sizes are significantly reduced, allowing for efficient storage without compromising access speed.

Elfs provides an innovative solution to the problem of efficient data storage and retrieval in modern computing

environments. By combining Zstandard compression with a real-time daemon system, Elfs ensures that users can interact with compressed data seamlessly, without experiencing any performance degradation. The file system efficiently manages both hot and cold data, ensuring that frequently accessed files remain uncompressed for quick access, while rarely accessed files are compressed to save storage space. With its focus on real-time performance and high compression ratios, Elfs stands as a strong candidate for modern file systems, especially in cloud and distributed storage environments.

Future work could explore additional features such as:

- a) Support for multi-threaded compression for further optimization.
- b) Enhanced error handling and fault tolerance in distributed environments.
- c) Extending compatibility with other file system formats and cloud services.

By leveraging Rust and advanced compression techniques, Elfs is well-positioned to meet the demands of modern data storage.

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APPENDIX

Appendix A: Technical Specifications

- Detailed system requirements
- Configuration parameters
- Performance metrics
- Benchmark results

Appendix B: Implementation Guidelines

- Installation procedures
- Configuration examples
- Troubleshooting guides
- Best practices

Appendix C: Performance Data

- Benchmark methodology
- Test results
- Comparative analysis
- Statistical evaluation

AUTHOR'S NOTE

This comprehensive research paper presents a detailed analysis of the ELFS file system, its implementation, and its impact on modern storage solutions. While every effort has been made to ensure accuracy, all technical specifications and performance metrics should be independently verified through practical implementation and testing.

